Criticality as a Resource: A Threshold Law for Analog Memory in Recurrent Networks, Its Boundaries, and an Architecture That Cannot Drift

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Abstract

We present an empirical study, comprising roughly sixty controlled experiments, of the relationship between dynamical criticality and memory in small recurrent networks. We establish a threshold law: retention of analog (continuous-valued) information requires the largest Lyapunov exponent of the recurrent dynamics to remain above $\lambda_{\max} \approx -0.08$, a condition that holds without exception across four mechanistically independent routes to subcriticality ($n = 40$, Spearman $\rho = 0.77$, $p = 5.6 \times 10^{-9}$). We show this law is exceptionless only within its proper domain: discrete (binary) information is retained instead through attractor dynamics, where contraction *protects* rather than destroys information, and computation itself requires no criticality at all (a hand-constructed parity automaton executes correctly at $\lambda_{\max} \approx -3.5$). Diagnosing an empirically observed memory-horizon failure as $\lambda_{\max}$-drift under long rollouts, we design a heterogeneous architecture in which one subspace is criticality by algebraic construction (an orthogonal "carrier" with all exponents exactly zero) rather than by training pressure. This extends the empirical memory horizon from roughly 80 to 800 timesteps, a result that survives a parameter-matched control against the original architecture, which performs *worse* when scaled to equal capacity. We report an extensive negative-results program: three independent structural mechanisms proposed to enable algorithmic discovery in the presence of sparse reward all fail identically, indicating the relevant bottleneck is supervisory rather than architectural. We further report a methodological finding of independent interest: an evaluation protocol that produced an apparently stochastic 33% success rate on an algorithm-discovery task was itself defective, conflating true task failure with an artifact of unclamped long-horizon rollout; correcting the evaluation instrument revealed a 100% underlying success rate, forcing a formal re-scoring of eight prior experimental conclusions. We conclude with a minimal mathematical model of the rollout-failure mechanism based on the polar decomposition of the hidden state into magnitude and direction, and report its validated and refuted predictions, including a demonstration that the identified failure mode does *not* generalize beyond a narrow task–architecture pairing. We make no claims regarding general intelligence or emergent computation; the contribution is a scoped, falsifiable, and heavily self-corrected account of one narrow mechanism.

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1 Introduction

This project began from a four-layer hypothesis, stated plainly so that its partial failure can be assessed honestly. The hypothesis was that a small *latent core* could generate the weights of a recurrent network at runtime (Layer 1); that a propagation mechanism could spread adaptation through the network's connectivity (Layer 2, "spider-web propagation"); that the network's dynamics

could be held at a self-organized critical point, $\lambda_{\max} \approx 0$, the so-called "edge of chaos" (Layer 3); and that intelligent, emergent behavior would arise from the interaction of the three (Layer 4).

Layer 1 survived essentially intact: a latent vector of a few dozen parameters, passed through a small generator network, can produce the recurrent weight matrix of a working sequence model. Layer 2 did not survive first contact with a gradient check: the propagation path was found to be structurally detached from the loss and contributed nothing (Section 10). Layer 3 survived, but in a form narrower and more precisely bounded than originally conceived: criticality is necessary for one specific computational demand (carrying continuous-valued information across time) and irrelevant or actively counter-indicated for others. Layer 4, emergent intelligence, did not appear, and nothing in the results below should be read as evidence toward it.

What replaced the original four-layer vision is a body of falsifiable, mechanistically grounded claims about recurrent dynamics, obtained by a discipline of pre-registered predictions, adversarial controls, and — on several occasions — the discovery that our own measurement instruments, not the phenomena under study, were the source of an apparent effect. We consider this last category of finding (Section 8) to be among the most valuable in the paper, independent of its bearing on criticality specifically.

Contributions. We claim four things precisely so that scope is unambiguous.

1. An exceptionless threshold law for analog memory retention as a function of the largest Lyapunov exponent, and its explicit domain boundary (Sections 3–4).
2. A structural fix, informed by the diagnosed failure mode, that replaces *maintained* criticality (a training-time equilibrium that can drift) with *constructed* criticality (an algebraic invariant that cannot), yielding a parameter-matched 10× improvement in memory horizon (Section 7).
3. A negative-results program showing that three structurally distinct mechanisms for algorithmic discovery all fail identically, localizing the bottleneck to the supervision signal rather than the architecture (Section 5).
4. A case study in the discovery and correction of an evaluation artifact, together with a minimal mathematical model of the underlying rollout instability and an honest report of that model's failure to generalize beyond its anchor case (Sections 8–9).

We do not claim to outperform any specific published architecture on any benchmark; no such comparison was run under conditions we would consider fair (see the discussion of tuning-artifact confounds in Section 12). We claim instead a set of scoped mechanistic findings, obtained under a self-imposed discipline of falsification, and we report our failures alongside our successes because in several cases the failures are the more informative result.

2 Methods

2.1 Model family

All experiments use small recurrent networks (hidden dimension typically 24–32) whose weights are, in the "reactor" variants, generated by a latent code z passed through a shallow nonlinear generator $g_\theta(z, x_t) \to W_t$, rather than stored as free parameters directly. Three recurrent update families are used, in increasing order of expressivity:

$$\text{Additive (A0):} \quad h_{t+1} = (1-\alpha)h_t + \alpha \tanh(W_h h_t + W_x x_t) \tag{1}$$

$$\text{Multiplicative (M2):} \quad h_{t+1} = (1-\alpha)h_t + \alpha \tanh\left((W_h + x_t W_m)h_t + W_x x_t\right) \tag{2}$$

$$\text{Heterogeneous (H):} \quad h_t = [c_t; a_t], \quad \text{Eqs. (6)–(7)} \tag{3}$$

Eq. (1) can only *add* the input's influence to the state; Eq. (2) allows the input to *multiply* the state, which we show in Section 4 is required for certain discrete computations.

2.2 Lyapunov spectrum estimation

The largest Lyapunov exponent $\lambda_{\max}$ and, where the full spectrum is needed, all d exponents $\lambda_1 \geq \cdots \geq \lambda_d$, are estimated via the Benettin algorithm with QR re-orthogonalization: an orthonormal frame is propagated through the Jacobians of the recurrence, QR-decomposed at each step, and the exponents recovered from the accumulated logarithm of the diagonal of R. The estimator was validated against systems with a closed-form spectrum (linear diagonal systems) to a maximum error below 10^{-2} before use on any trained model.

2.3 Rigor infrastructure

Three pieces of shared infrastructure were built in direct response to artifacts discovered during the project and are used throughout: a TrajectoryLogger that records the loss, accuracy, $\lambda_{\max}$, and gradient norm at every epoch (rather than only at the final epoch, which was found to conceal several artifacts described below); a MixedDelaySampler that trains on a distribution of task difficulties rather than a single fixed difficulty, with held-out evaluation reported separately for interpolation and extrapolation; and a FairBaselineHarness that applies an identical hyperparameter search budget to every compared architecture. A companion verdict-generation library, described in Section 11, enforces that every reported conclusion is selected programmatically from measured quantities rather than written in advance.

3 The Threshold Law

Law 1 (Analog memory threshold)**.** *Let $\lambda_{\max}$ denote the largest Lyapunov exponent of a recurrent network's trained dynamics. Retention of a continuous-valued (analog) signal across a task-relevant time horizon requires*

$$\lambda_{\max} \gtrsim -0.08. \tag{4}$$

This condition is necessary but not sufficient: within the band $\lambda_{\max} \in [-0.08, +0.05]$, the value of $\lambda_{\max}$ does not predict the achievable memory horizon.

Law 1 was established across four independent means of varying $\lambda_{\max}$: a homeostatic loss term $\gamma(\lambda_{\max} - \lambda^*)^2$ with varying target λ^*; direct variation of the recurrent weight initialization scale; variation of the leak rate α; and variation of the generator's output gain. Across $n = 40$ trained models spanning these four routes, the achievable memory horizon and $\lambda_{\max}$ are related by Spearman $\rho = 0.77$ ($p = 5.6 \times 10^{-9}$), and the relationship is a hard threshold rather than a smooth trend: every model with $\lambda_{\max} < -0.08$ achieved zero memory retention, and every model with $\lambda_{\max} > -0.08$ achieved a nonzero, though variable, retention horizon.

Finding 1 (Self-organization)**.** *A homeostatic loss $\gamma\lambda_{\max}^2$ drives networks initialized away from criticality, in either direction, back toward $\lambda_{\max} \approx 0$ during training. Networks initialized subcritical ($\lambda_{\max} = -0.19$) and supercritical ($\lambda_{\max} = +0.04$) both converge to $\lambda_{\max} \in [-0.01, -0.003]$ under this loss, and the resulting networks solve a memory task (delay $= 80$) that the same architecture without the homeostatic term fails at (59.5% vs. 100% accuracy in the best configuration tested).*

We note one important scope correction to Finding 1, obtained later in the project (Section 8): the claim that $\lambda_{\max}$-band residence is *necessary* for algorithmic acquisition was subsequently shown

to conflate the metric used to score acquisition with an unrelated rollout artifact. The corrected claim is that $\lambda_{\max}$ governs *rollout stability under long unclamped evaluation*, not whether a task's underlying algorithm was learned. We flag this here to avoid the reader's carrying forward Finding 1 in its original, uncorrected form.

4 The Storage Dichotomy

Law 1 governs one specific kind of information, and testing its boundary was as important as establishing it.

Finding 2 (Computation requires no criticality)**.** *A hand-constructed recurrent module implementing the running-parity automaton via a strong multiplicative gate (Eq. (2) with $\alpha = 1$, $W_x = 0$, $W_h = 0$, $W_m = cI$ for large c) computes parity correctly at every timestep while operating at $\lambda_{\max} \approx -3.49$ (a fully flat spectrum: all measured exponents equal to within numerical precision). Criticality is not required to* transform *information; the threshold law in Section 3 governs only the passive* carrying *of continuous-valued information forward in time.*

Finding 3 (Discrete information escapes via attractor storage)**.** *A network solving a binary (as opposed to continuous-valued) retention task operates at $\lambda_{\max} \approx -0.42$ — well below the analog threshold of Law 1 — and nonetheless retains its stored bit perfectly. A mid-sequence state perturbation of magnitude ϵ* decays *back toward the correct stored value in this regime (attractor storage), whereas an equivalent perturbation applied to a network storing continuous-valued content* persists *and corrupts the stored value (analog carrying). Contraction protects discrete information rather than destroying it, because the perturbed state is drawn back into the same discrete attractor basin that encodes the correct answer.*

This dichotomy — analog information requires $\lambda_{\max} \gtrsim -0.08$; discrete information is protected by $\lambda_{\max}$ well below that threshold via attractor dynamics; computation requires neither — resolves what would otherwise appear as a direct contradiction between Finding 2/3 and Law 1, and explains several apparent exceptions to the threshold law observed in earlier, less controlled experiments (a subcritical model retaining one bit over a short horizon, for instance, is using attractor storage rather than violating the analog law).

5 Learnability of Computation Under Sparse Supervision

5.1 The discovery wall

Training the recurrent parity automaton from a random initialization, with only a final-step loss and no curriculum, fails to reach above-chance accuracy in every configuration tested (additive, multiplicative, and heterogeneous architectures; 0/9 seeds across variants), with the loss pinned at $\ln 2$ (the entropy of a uniform binary guess) for the full training budget. This is a *learnability* failure rather than an *expressivity* failure: a network hand-initialized at the known correct solution remains at 100% accuracy under continued gradient descent (the solution is a stable attractor of the optimizer), which rules out the possibility that the correct function is simply unreachable in the model's function class.

5.2 Three falsified remedies

We tested three structurally distinct mechanisms intended to cross this discovery wall, none of which involve giving the model additional information about the task's intermediate structure.

- **Flow.** A carrier subspace constructed to be an exact gradient highway (orthogonal recurrence, Section 7) delivers a gradient to the earliest timestep roughly $10^5\times$ larger than the baseline architecture. Parity accuracy remains at chance (0/3 seeds).
- **Information.** An auxiliary loss requiring the network to reconstruct its last $k = 8$ inputs from its current state drives reconstruction error to 2×10^{-4} (near-perfect), while parity accuracy remains indistinguishable from the no-auxiliary-loss control (0/3 seeds, best accuracy 0.555 vs. control 0.539).
- **Composition.** A library of small learned modules, pre-trained on a short version of the task and then frozen while a router is trained to compose them on the full-length task, also fails (0/3 seeds), and post-hoc analysis shows the pre-trained "atom" was never a general step rule but a length-specific static lookup (a precondition failure discovered and repaired mid-experiment).

Finding 4 (The discovery wall is supervisory, not architectural)**.** *Across all three mechanisms, and across every architecture tested, the only interventions that reliably cross the discovery wall are ones that alter the* supervision signal itself*: a length curriculum (starting at a trivially short sequence and increasing on success) solves the task in a median of four epochs; a homeostatic $\lambda_{\max}$-control loss also solves it, but 185× more slowly (median 742 epochs), because it acts as an indirect proxy for gradient flow rather than directly reshaping the loss landscape. No architectural modification we tested substitutes for either.*

5.3 A falsified schedule: "critical to search, contract to commit"

Motivated by the observation that the parity automaton's stable solution operates at $\lambda_{\max} \approx -3.3$ (Finding 2) while the homeostatic-loss route to discovery requires $\lambda_{\max} \approx 0$ throughout training, we tested whether *annealing* the target from 0 to a subcritical value during training could combine the fast discovery of criticality with the natural operating point of the solution. Both a linear and a discovery-triggered annealing schedule failed to solve the task at any tested configuration (0/9 across three schedule variants), while a control that clamped $\lambda_{\max}$ to 0 for the entire training run succeeded (3/3). A trajectory-level diagnostic showed that annealing schedules which began contracting even shortly before the point at which the fixed-$\lambda_{\max} = 0$ control would have succeeded (in one case, 53 epochs prior) irrecoverably arrested discovery. We conclude that criticality must be maintained for the *duration* of the discovery process rather than the duration of training, and that contraction into the final solution is a byproduct of task convergence rather than a schedule that can be externally imposed.

6 Selective Criticality: A Controllable Mechanism Without Demonstrated Utility

Given that gated architectures admit an approximate identity linking the trained gate value g_i of dimension i to that dimension's local Lyapunov exponent,

$$\lambda_i \approx \log g_i, \tag{5}$$

verified both analytically (maximum error 0.005 against a closed-form derivation) and empirically in trained networks (Pearson $r = 0.893$), we asked whether the *full spectrum* — not merely $\lambda_{\max}$ — could be deliberately shaped. A loss term targeting a specific number K of near-zero exponents, with the remainder targeted toward a contractive value, reliably produces exactly K measured near-zero

exponents (slope of realized vs. target channel count = 1.17; seed-to-seed standard deviation = 0.00 at $K = 2$, versus 1.63 for a naive scalar-only homeostatic loss targeting only $\lambda_{\max}$).

We then asked the harder question: does this controllability confer any *performance* benefit over a network left to self-organize its own spectrum. Tested across four qualitatively distinct regimes — abundant representational capacity, scarce capacity (below the information-theoretic minimum), tight but sufficient capacity, and a task whose storage/computation demand varies unpredictably across episodes — the answer was uniformly negative. In every regime, an unshaped network matched or exceeded a network with an explicitly shaped spectrum, and the shaped network incurred additional training cost with no compensating benefit. In one regime (variable per-episode demand), an initial apparent advantage for explicit shaping was traced, under a hardened re-training sweep, to the unshaped baseline having been undertrained at a single learning rate; properly trained, the unshaped network matched the shaped one exactly.

Finding 5 (Self-organization is sufficient)**.** *Across every regime we tested — representational abundance, scarcity, tight sufficiency, and unpredictable variable demand — gradient descent already discovers whatever spectral allocation an explicit control mechanism would otherwise impose. Selective criticality is a valid, precisely controllable* characterization *tool (Eq.* (5)*) and the shaping result above are both real and reproducible) but we found no evidence that it is a useful* performance *tool.*

7 A Structural Fix: Criticality by Construction

7.1 Diagnosis

The original reactor architecture, trained on a delayed-recall task, solves delays up to approximately 60–80 steps and fails beyond that point. Measuring $\lambda_{\max}$ across training reveals that networks solving the task operate near $\lambda_{\max} \approx 0$ at short delays but drift toward $\lambda_{\max} \approx -0.15$ as the required delay approaches and exceeds the point of failure. Because $\lambda_{\max}$ under the standard architecture is a property of a trained equilibrium (maintained by the interaction of the loss and the recurrent weights), rather than an algebraic invariant, this equilibrium can and does drift under sufficient task pressure.

7.2 A carrier subspace with criticality by construction

We propose splitting the hidden state into two coupled subspaces, $h_t = [c_t; a_t]$, with distinct dynamical roles:

$$c_{t+1} = R\, c_t + W_{\text{in}}(z, x_t)\, x_t + P_{ac}\, a_t, \tag{6}$$

$$a_{t+1} = g_t \odot a_t + (1 - g_t) \odot \tanh\big(W(z, x_t)\, a_t + U x_t\big) + P_{ca}\, c_t, \tag{7}$$

where R is constrained to the orthogonal group (parameterized via the Cayley transform, $R = (I - A)^{-1}(I + A)$ with A skew-symmetric), and P_{ac}, P_{ca} are learned coupling projections. Because $R^\top R = I$ exactly, every Lyapunov exponent of the carrier subspace in isolation is exactly zero *by algebraic construction*: there is no training pressure that can move it, because there is no parameterization of R under which it is not orthogonal.

We verified this by construction rather than assuming it: measured orthogonality error of the trained R was 2.4×10^{-7}; the measured carrier-subspace spectrum after training had maximum $|\lambda_i| < 10^{-7}$ (versus a 10^{-2} threshold used elsewhere in this work to validate the spectrum estimator itself); and a loss injected at timestep $t = 60$ produced a gradient at $t = 0$ within a factor of 1.0000 of its magnitude at $t = 60$ — a lossless gradient highway across the full rollout.

7.3 Result and control

Finding 6 (Horizon extension, parameter-matched)**.** *The heterogeneous architecture (Eqs.* (6)–(7)*) solves the delayed-recall task at delays of* 80*,* 120*,* 200*, and* 400 *steps at* 3/3 *seeds each (accuracy* ≥ 0.999*), and at* 1/2 *seeds at delay* 800*. Critically, this is not attributable to the additional parameters introduced by the two-subspace design: the original architecture, scaled to match the heterogeneous model's parameter count (*8461 *vs.* 8530*),* still fails *at delay* 80 *(*0/3 *seeds, best accuracy* 0.54*) and drifts to a* more *subcritical* $\lambda_{\max}$ *than the smaller, unscaled original (*-0.15 *vs.* -0.02 *at delay* 200*). Additional representational capacity, absent the algebraic constraint, deepens the drift rather than correcting it.*

The residual failure at delay 800 was traced, via a subsequent measurement thread, not to renewed $\lambda_{\max}$-drift (the carrier spectrum remains exactly zero, $\max|\lambda_i| < 10^{-7}$, even at this horizon) but to a scale mismatch between the trained linear readout and the carrier's operating radius at extreme horizons; a closed-form readout refit at the correct scale, combined with a state rescaling, recovers roughly 29 percentage points of the deficit, though not to the 99% threshold used elsewhere in this work. We report this honestly as a partially, not fully, resolved failure mode.

7.4 Relation to prior architectures

Recurrent networks with an orthogonal or norm-preserving state transition are an established architectural family (unitary and orthogonal RNNs; see Section 12), and we do not claim the carrier subspace's algebraic construction is itself novel. Our contribution here is the *diagnostic path* to this design — measuring $\lambda_{\max}$-drift as the specific cause of an empirical failure before selecting orthogonality as its targeted cure — and the explicit two-subspace coupling that lets a norm-preserving memory channel coexist with a saturating, non-norm-preserving computation channel in the same network, which we use in Section 5 to dissociate flow-limited from landscape-limited learning failures within one architecture.

8 An Evaluation Artifact and Its Correction

We report this section in some methodological detail because we believe the general lesson — that a metric which discriminates between experimental conditions is not thereby validated as measuring the intended construct — is of independent interest.

8.1 The apparent phenomenon

Training the parity-discovery task under the curriculum regime described in Section 5 produces a network that solves every curriculum stage, including the full task length, at $\geq 99\%$ accuracy. When such a network is subsequently evaluated *zero-shot* at a much longer sequence length (an unclamped rollout to $T = 200$, versus a training length of $T = 20$), only approximately 33% of seeds (2/6) retain high accuracy; the remainder collapse to chance. This appeared, across an extensive follow-up program (Sections omitted here for space; see project repository), to be a genuinely stochastic phenomenon: seed-dependent, resistant to weight decay, resistant to a mixed-length training curriculum, resistant to an explicit $\lambda_{\max}$-homeostatic loss, and traceable via a checkpoint-fork-and-replay design to noise in the stochastic gradient trajectory itself rather than to the network's initialization.

8.2 The correction

A subsequent, unrelated investigation (into an apparent long-horizon memory failure of the heterogeneous architecture at delay 800; see Section 7) established that an inference-time radius clamp — rescaling the hidden state at each rollout step to match the network's own typical trained-length norm, with no retraining whatsoever — could recover long-horizon accuracy that an unclamped rollout reported as failed. Applying the identical clamp, with no retraining, to the "33%" parity-discovery models produced 6/6 seeds at ceiling accuracy, including the seeds that had previously read as complete failures (accuracy $\to 1.0$ from a pre-clamp value near chance).

Finding 7 (The metric, not the model, was the source of variance)**.** *Every one of the six seeds evaluated had, in fact, learned the correct parity algorithm. The apparently stochastic* 33% *success rate reported across dozens of subsequent experiments was an artifact of an unclamped long-horizon evaluation protocol conflating true task failure with a state-magnitude drift that the (correctly learned) recurrence and the (correctly trained) readout had simply never been exposed to during training.*

A validity control accompanies this finding: an *untrained* network, evaluated under the identical clamp, achieves at most 0.514 accuracy (chance), confirming that the clamp cannot manufacture a correct algorithm where none exists; the rescue is specific to networks that had already learned the task.

8.3 Re-scoring

This correction required a formal re-scoring of every prior experimental conclusion that had used the unclamped long-horizon evaluation as its scoring criterion. Of eight affected conclusions, five were revised outright (including the claim, reported in Section 3, that $\lambda_{\max}$-band residence is necessary for algorithmic *acquisition*; the corrected claim is that it is necessary only for rollout *robustness* under the specific evaluation protocol used), one was partially revised, one (an unrelated horizon-fraying attribution) stood unchanged after the clamp failed to rescue it, and one was reframed as a special case of the newly identified artifact rather than an independent training-time effect. We regard the discipline that produced this correction — adding an explicit control proving a metric *can* fail before trusting that it discriminates for the intended reason — as itself a contribution, and we return to it as a methodological recommendation in Section 13.

9 A Minimal Model of the Rollout Instability

9.1 The model

Writing the hidden state in polar form, $h = ru$ with $r = \|h\|$ and $u = h/\|h\|$, we hypothesized a minimal model in which the task-relevant code is carried in the direction u, while the magnitude r evolves under an approximately autonomous radial map,

$$r_{t+1} \approx g(r_t)\, r_t, \tag{8}$$

whose stability, rather than the code itself, determines whether a long rollout remains within the distribution the readout was trained on.

9.2 Supporting measurements

Across twelve (task, architecture) cells (the parity-discovery anchor case together with a token-copy task and a modular-sum task, each on the heterogeneous architecture, a gated recurrent unit, and a

long short-term memory unit), we found: task information decodes from the direction u nearly as well as from the full state h in eleven of twelve cells, while decoding from magnitude alone never exceeds chance; the realized radial gain, $g_{\text{real}} = \|h_{200}\|/\|h_{20}\|$ per step, exceeds 1 in every cell that fails at long rollout and is within 1% of 1 in every cell that succeeds; and the affine readout's decision boundary is empirically insensitive to state rescaling (a bias-flip analysis attributes at most 6% of the deficit to scale-sensitivity of the readout itself, with the remainder attributable to the state's direction having genuinely left the region on which the readout was calibrated).

9.3 Where the model succeeds and where it does not

The radial model correctly explains, post hoc, why the parity-discovery anchor case is rescuable by a pure magnitude clamp (its code is directional and its failure mode is magnitude runaway) and why a token-identity code on the same architecture is not (the information remains linearly present, per a closed-form decoding probe, but the trained *readout* rather than the state's radius is misaligned; a readout refit, not a clamp, recovers it). We then tested the model's generality directly, across a 12-task-by-5-architecture grid spanning two axes the model predicts should matter (supervision density and the presence or absence of an algebraic norm bound on the recurrence).

Finding 8 (The rollout phenomenon does not generalize)**.** *Of five pre-registered predictions derived from the radial model and tested out of sample, one held cleanly (a gated recurrent unit remains free of magnitude drift regardless of supervision density, replicating an earlier finding), one was confounded by an unrelated learnability limitation of the specific architecture used to test it, and three were falsified: magnitude drift is not confined to sparse supervision (two densely supervised seeds on the anchor task also drifted), the magnitude clamp is not effective only for directionally coded tasks (it rescued several non-directional cells), and readout misalignment is not confined to identity-coded tasks. The rollout-failure phenomenon that motivated Section 8 is real and reproducible, but it is confined to the specific task–architecture pairing on which it was originally discovered, and does not constitute a general law of recurrent execution.*

We consider this an important, if deflationary, result. A subsequent attempt to unify the surviving anomalies under a single "state-distribution excursion" measure, using a leave-one-cell-out validation against the existing experimental grid rather than a new anchor case, found that the proposed measure reduces to the magnitude ratio already in Eq. (8) wherever the two are jointly measurable (Pearson correlation 0.99 between the two quantities; identical predictive accuracy), and diverges from it in exactly one case — a memory-horizon failure at extreme delay (Section 7) that manifests as state *collapse* rather than *runaway* — indicating that the execution-time failure family comprises at least two distinct mechanisms with opposite sign, not one axis with a single governing scalar.

10 The Falsification Record

We list every claim in this project that was pre-registered or seriously entertained and subsequently falsified by a controlled experiment, because we consider the discipline of reporting these to be inseparable from the credibility of the claims that survived.

1. **Spider-web propagation** (original Layer 2). Found to be gradient-detached from the loss; ablation across the propagation coefficient produced no measurable effect.
2. **Dynamic latent-code conditioning.** A frozen latent code performed identically to a dynamically updated one, indicating the dynamic update contributed nothing.

3. **A smooth Lyapunov–memory law.** Superseded by the threshold law (Section 3); the relationship is a hard threshold, not a continuous trend.
4. **Architectural superiority over standard gated recurrent networks.** An initial result favoring the reactor architecture was traced to an undertuned baseline (a single, too-low learning rate); a symmetric hyperparameter search showed properly tuned gated recurrent units matching the reactor on the original task.
5. **General-purpose memory-horizon extension via the homeostatic loss.** The apparent benefit did not generalize beyond the specific delay at which the network was trained.
6. **"Critical to search, contract to commit" scheduling** (Section 5).
7. **A stable split spectrum as a general law.** An initially observed one-critical-channel-plus-contractive-remainder spectrum, obtained under the homeostatic loss, was later shown to be seed-fragile and specific to a training regime with an extended loss plateau, not a general property of solved instances of the task.
8. **K discrete bits require K critical channels.** Discrete information is instead stored via attractor dynamics (Section 4); channel count did not scale with bit count.
9. **Selective criticality confers a performance advantage** (Section 6), tested and falsified in four independent regimes.
10. **Three structural remedies for the discovery wall** (flow, information, composition; Section 5).
11. **Radial excursion as a unifying execution-time principle** (Section 9), found to reduce to the simpler magnitude-ratio measure wherever jointly measurable, and to reverse sign on an independently discovered failure mode.

11 Verdict Discipline

A recurring failure mode in this project's own experimental software, caught on at least three separate occasions, was the pre-writing of a conclusion's text before the branching logic that was meant to select it had been fully specified, such that a template string reflecting the *expected* outcome was emitted regardless of the measured result. This is, in miniature, the same failure as Section 8 at the level of code rather than experimental protocol: a reporting instrument silently substituting an assumption for a measurement. In response, all subsequent experiments in this project generate their reported conclusions exclusively through an explicit branch-selection library, every branch of which is exercised under a synthetic-data dry run before being trusted against real results, together with a lint pass that flags conclusion text contradicting the specific measured values it purports to summarize (for instance, text describing a decoding accuracy as "chance" when the measured value exceeds a stated chance threshold). We recommend this discipline, or an equivalent, for any research program in which conclusions are generated programmatically from a large and evolving experimental codebase.

12 Related Work

Edge-of-chaos and reservoir computing. The hypothesis that computational capacity in recurrent systems is maximized near a critical point separating ordered and chaotic dynamics has a long history in reservoir computing and echo-state networks. Our threshold law is consistent with this tradition for the specific case of analog memory, but our results indicate the relationship is a hard threshold rather than a smoothly peaked capacity curve, and that it does not extend to computation, contrary to some framings of the "edge of chaos" hypothesis as a general computational principle.

Orthogonal and unitary recurrent networks. The use of norm-preserving (orthogonal or unitary) recurrent transition matrices to combat vanishing and exploding gradients over long sequences is an established technique. We do not claim priority on the algebraic construction used in our carrier subspace (Section 7); our contribution is the diagnostic route to this design via explicit Lyapunov-spectrum measurement, and the coupling of a norm-preserving subspace with a separate, saturating computation subspace within one architecture, which we use to empirically dissociate flow-limited from landscape-limited learning failure (Section 5).

Gating as dynamical control. The approximate correspondence between a trained gate value and a local stability exponent (Eq. (5)) is consistent with existing analyses of gated architectures as learned, per-dimension timescale or eigenvalue controllers. Our contribution is the explicit measurement of this correspondence via the full Lyapunov spectrum and the subsequent (negative) test of whether deliberately shaping that spectrum, rather than leaving it to self-organize, confers a performance benefit.

Exposure bias and compounding error. The rollout-instability phenomenon in Section 8 bears a family resemblance to the general problem of compounding error under closed-loop (self-conditioned) evaluation, a concern raised in the sequence-generation literature under headings including exposure bias and addressed by techniques such as scheduled sampling. We did not set out to study this problem and arrived at a narrow instance of it via an unrelated architectural investigation; we believe the specific diagnostic value here is the demonstration that the resulting metric failure can silently drive an extended, otherwise well-controlled experimental program to an incorrect conclusion for months before an unrelated investigation exposes it.

Library learning and modular composition. Our falsified "composition" remedy for the discovery wall (Section 5) instantiates ideas present in modular meta-learning and program-induction approaches to compositional generalization; we do not claim novelty in the architectural concept, only in the specific negative result and its diagnosis (the learned solution's input-conditioning was distributed across the router-library pair rather than localized to a single reusable module, such that the freeze-and-reuse protocol discarded exactly the component carrying the input-dependent part of the solution).

We are not aware of a prior report of the specific evaluation artifact in Section 8 (a metric conflating algorithmic non-acquisition with an unrelated rollout-magnitude drift) in this experimental setting, though we would not be surprised if the general phenomenon has been observed, informally or formally, elsewhere.

13 Limitations

- All experiments use small networks (hidden dimension on the order of 10–30) on synthetic sequence tasks (delayed recall, running parity, token copy, modular arithmetic). No claim in this paper has been tested at a scale, or on data, resembling a practical application, and we regard the question of whether the threshold law (Section 3) holds recognizably in larger, densely trained models (for instance, small recurrent language models) as the single most important open question raised by this work.
- The precise width of the transition band around $\lambda_{\max} \approx -0.08$ was estimated from $n = 40$ models across four routes to subcriticality but was not subjected to a dedicated high-resolution sweep; we report it as a band (roughly -0.08 to -0.10) rather than a single value, and a borderline case (a tuned long short-term memory network at $\lambda_{\max} = -0.098$ solving its task) suggests the boundary may be somewhat softer, or somewhat architecture-dependent, than a single scalar threshold implies.

- The generality analysis in Section 9 covers a 12-cell grid; several architectures of clear relevance (state-space models with an enforced stability parameterization, of the kind used in several current architectures, and attention-based models, for which the polar decomposition's underlying object, a single evolving recurrent state, does not straightforwardly apply) were deliberately excluded rather than tested with an under-resourced implementation, and we regard both as important, currently open, extensions.
- Several quantitative comparisons in this paper (Section 10, item 4) are explicitly reported as *failed* attempts at architectural comparison, precisely because an initial comparison was found to rest on an unfair baseline. We regard this as a strength of the paper's discipline rather than a gap, but we note explicitly that no fair, published-baseline comparison against a standard architecture on a standard benchmark is offered here.
- The residual failure mode at extreme memory horizon (delay 800, Section 7) is only partially explained (a readout-refit recovers part, not all, of the deficit); we did not identify a complete mechanistic account.

14 Future Work

- **Scale.** Measuring the full Lyapunov spectrum of a small recurrent language model, trained densely on real text, to test whether the threshold law and the storage dichotomy (Sections 3–4) are properties of small synthetic tasks or survive to a setting with genuine long-range analog and discrete content mixed together. This requires first resolving a known numerical instability (NaN propagation through the Benettin estimator when applied to gated architectures with saturating internal state), parked but unresolved in the present work.
- **A dense, task-agnostic auxiliary signal for the discovery wall.** All three falsified remedies in Section 5 avoided giving the model information about intermediate task structure; a natural, still-untested candidate is a per-step self-supervised objective aligned with the task's own recursive structure (as opposed to the task-agnostic history-reconstruction objective we tested and falsified), which would need to be shown to generalize across tasks to be more than a restatement of the curriculum result.
- **Per-dimension stability parameterizations across the modern state-space model family.** The gate–exponent correspondence (Eq. (5)) and the algebraic-orthogonality construction (Section 7) both suggest a unified account of why several current linear-recurrence architectures enforce an explicit stability constraint on their decay parameter; a controlled, matched-pair comparison (an unconstrained diagonal linear recurrence against an otherwise identical one with an enforced $|a| \leq 1$ parameterization) was attempted in this work but was confounded by a learnability limitation of the specific task chosen, and remains open.
- **A complete account of the extreme-horizon readout failure.** Section 7's residual delay-800 deficit was partially, not fully, explained; a full account would likely require separating contributions from finite numerical precision, readout capacity, and any residual coupling-induced instability between the carrier and computation subspaces.

15 Conclusion

We set out to test whether dynamical criticality is a general principle of intelligent computation in recurrent networks. It is not, by any evidence obtained in this work. What survived a sustained internal effort to falsify it is smaller and more precisely bounded: criticality is necessary, and in our tests sufficient, for one specific computational demand — carrying continuous-valued information forward across time — and irrelevant or counter-indicated for others. We were able to convert this

narrow, measured diagnosis into a working architectural improvement (Section 7) by replacing a trained equilibrium with an algebraic invariant, and we were able, in the course of an unrelated investigation, to discover and correct a defect in our own evaluation methodology that had silently driven an extended experimental program to an incorrect conclusion (Section 8). We regard the totality of what we did *not* find — three falsified remedies for algorithmic discovery, a falsified general law of rollout instability, a falsified performance benefit for controllable criticality, and an extended falsification record spanning the original four-layer hypothesis this project began from — as inseparable from the credibility of what we did.

Code, experiment logs, and figures referenced in this paper are maintained in a private repository and available on request.